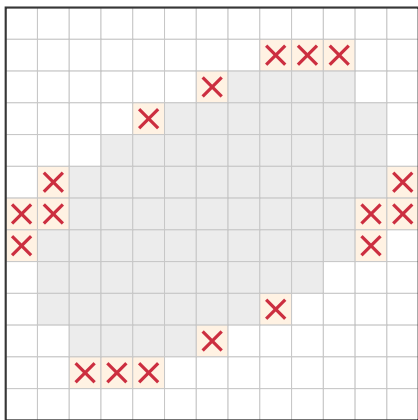


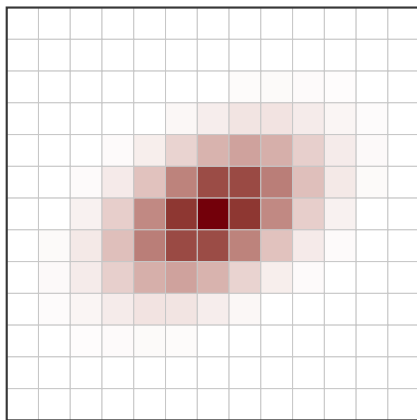
Elliptical Kernel Construction

(A) Mask M_E



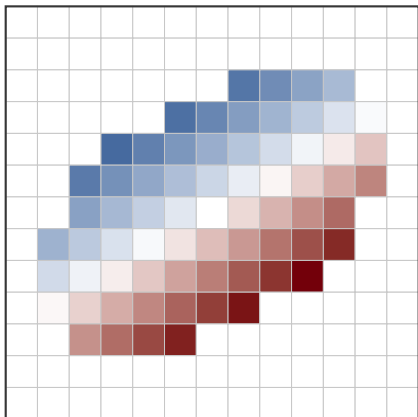
$$M_E(i, j) = \mathbb{1} \left[\frac{u^2}{\sigma_u^2} + \frac{v^2}{\sigma_v^2} \leq r_0^2 \right]$$

(B) Envelope $G(u, v)$

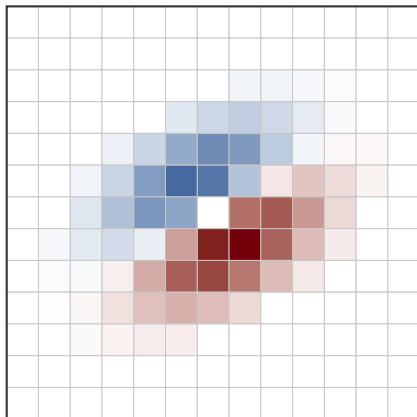


$$G(u, v) = \exp \left(-\frac{1}{2} \left(\frac{u^2}{\sigma_u^2} + \frac{v^2}{\sigma_v^2} \right) \right)$$

(C) Derivative $-v$



(D) Combined K_E



$$K_E = -v \cdot G(u, v) \cdot M_E$$

$-v$ profile: positive (garnet) / negative (blue)